

FinBERT vs FinGPT: Enhancing Malaysian Stock Predictions with News-Driven Insights

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Research Motivations

- Why this problem?
 - Malaysia's stock market influenced by political and financial news, but current models rarely account for this.
 - Sentiment analysis using FinBERT and FinGPT could help forecast market movements more accurately.
- Challenges:
 - Lack of Malaysian-focused sentiment models.
 - Integration of news sentiment with stock time-series data.
 - FinBERT is pre-trained on US data while FinGPT can be fine-tuned but needs testing.

Industry Background and Publications

- **Industry background**
 - Financial sentiment analysis is gaining traction in algo-trading.
 - News-driven models are used by hedge funds, quant researchers, and fintech companies
- **Key publications**
 - Chuan, Tan Boon, Vazeerudeen Abdul Hameed, and Chandra Reka Ramachandiran. "Malaysia stock market prediction using artificial intelligence." *Journal of Applied Technology and Innovation* (e-ISSN: 2600-7304) 5.1 (2021): 32.
 - Kang, Ji-Won, and Sun-Yong Choi. "Comparative investigation of gpt and finbert's sentiment analysis performance in news across different sectors." *Electronics* 14.6 (2025): 1090.
 - jun Gu, Wen, et al. "Predicting stock prices with finbert-lstm: Integrating news sentiment analysis." *Proceedings of the 2024 8th International Conference on Cloud and Big Data Computing*. 2024.

Research Objectives

- Compare FinBERT and FinGPT on sentiment analysis of Malaysian news.
- Evaluate impact of sentiment on KLSE stock price movements (daily returns, volatility)
- Combine sentiment features with time-series data for price prediction.
- Build and benchmark multiple prediction models:
 - LSTM/GRU
 - Sentiment-enhanced models
 - FinGPT-augmented transformer models

Research Methodology & Structure

- **Exp 1: Sentiment Model Benchmarking**
 - Compare FinBERT and FinGPT in extracting sentiment from Malaysian financial and political news.
 - Evaluate model output on labelled samples (qualitative + quantitative)
- **Exp 2: Sentiment vs Stock Movement**
 - Investigate the effect of daily sentiment on KLSE stock returns and volatility
 - Conduct lag analysis: T, T+1, T+2 with regression & correlation metrics
- **Exp 3: Stock Prediction Models**
 - Build time-series prediction models with and without sentiment features
 - Compare baseline vs FinBERT-enhanced vs FinGPT-enhanced models.

Exp 1: Sentiment Model Benchmarking

Experiment

- Benchmark sentiment classification capability of FinBERT and FinGPT
- Use both zero-shot and fine-tuned versions
- Compare contextual relevance and polarity output

Data

- Curated Malaysian financial/political news headlines (scraped or collected from sources like The Star, NST, Malaysiakini)
- Small manually annotated subset (positive/negative/neutral sentiment)

Models

- FinBERT (zero-shot + optional fine-tune)
- FinGPT (financial LLM) with inference on same news

Results

- F1-score, precision, recall comparison
- Examples where FinGPT captures better context
- Visual: bar chart of accuracy comparison per class

Exp 2: Sentiment vs Stock Movement

Experiment

- Test correlation between daily sentiment score and stock returns (next-day, 2-day, etc.)
- Use both collected daily sentiment and stock movement

Models

- Pearson/Spearman correlation
- Linear regression
- Granger causality test

Data

- Daily sentiment scores from Exp 1 outputs
- Historical data from Yahoo Finance API for selected Malaysian top companies' stock

Results

- Correlation strength over different lags
- Visual: Line plot of sentiment vs stock return trend, lag correlation heatmap

Exp 3: Stock Price Prediction Models

Experiment

- Train and evaluate predictive models for stock prices using different inputs
- Perform feature engineering on historical stock data: moving average, RSI, MACD, daily returns, volatility
- Compare performance between models using no sentiment, FinBERT sentiment, FinGPT sentiment

Models

- LSTM or GRU baseline (price features only)
- LSTM/GRU + FinBERT sentiment + engineered price features
- LSTM/GRU + FinGPT + engineered price features

Data

- Historical stock data + engineered features
- Sentiment scores from Exp 1
- Train-test split

Results

- RMSE, MAE, directional accuracy for each model
- Best-performing model using FinGPT/FinBERT + prediction models + features
- Visual: Model comparison table and performance bar chart

Contributions to Science (academic)

- First Benchmarking of FinBERT vs FinGPT on Malaysian News
 - New study on comparing these two models for Malaysian financial and political contexts.
 - Contributes to research on domain adaptation and model transferability in NLP.
- Multimodal Financial Prediction Framework
 - Integrate sentiment-analysis with time-series forecasting.
 - Improve the study of hybrid models that combine language understanding with numerical prediction.
- Application of Transformer-based Models in Emerging Markets
 - Demonstrates how transformer models can be fine-tuned or adapted for non-Western financial markets.
 - Encourage further academic exploration in understudied economies like Malaysia
- Localized Sentiment Annotation and Analysis
 - Creation of labelled dataset of Malaysian financial news with sentiment tags.
 - Support future research in regional NLP.

Impact Statement (business/industry)

- Improved Trading Strategies for Local Investors
 - Traders can use sentiment-enhanced models to make more informed buy/sell decisions.
 - Helps in identifying market-moving news faster than traditional methods.
- Enabling Real-time Market Monitoring Tools
 - Supports development of AI-driven dashboards that summarise daily sentiment and market outlook.
 - Useful for retail investors, asset managers, and algorithmic trading desks in Malaysia.
- Lower Barrier to Entry for Quantitative Methods in ASEAN
 - Provides a working template for low-cost, data-driven prediction systems.
 - Encourage financial startups or SMEs to adopt AI/NLP solutions without heavy infrastructure.
- Advancing Financial Analytics in Malaysian Capital Markets
 - Encourage Bursa Malaysia and fintech firms to invest in AI-based analytics.
 - Potential to improve market transparency, investor confidence, and data-driven decision making.